StudySapiens Board MCQs

Chapter 3 — Cells and Subcellular Organelles

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The basic unit of structure and function in living organisms is the:

(a) Organ
(b) Cell
(c) Tissue
(d) Organelle

2. The term cellulae was coined by:

(a) Robert Hooke
(b) Rudolf Virchow
(c) Theodor Schwann
(d) Robert Brown

3. Living cells in pond water were first observed by:

(a) Anton van Leeuwenhoek
(b) Matthias Schleiden
(c) Louis Pasteur
(d) Robert Hooke

4. The nucleus was discovered by:

(a) Robert Brown
(b) Theodor Schwann
(c) Robert Hooke
(d) Rudolf Virchow

5. The statement that all plants are composed of cells was given by:

(a) Theodor Schwann
(b) Matthias Schleiden
(c) Rudolf Virchow
(d) Robert Hooke

6. The statement that all animals are composed of cells was given by:

(a) Matthias Schleiden
(b) Theodor Schwann
(c) Louis Pasteur
(d) Robert Brown

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7. The idea that all cells come from pre-existing cells was proposed by:

(a) Theodor Schwann
(b) Matthias Schleiden
(c) Robert Hooke
(d) Rudolf Virchow

8. Which of the following is an exception to cell theory?

(a) Protists
(b) Fungi
(c) Viruses
(d) Bacteria

9. Mitochondria and chloroplasts challenge cell theory because they:

(a) Lack membranes
(b) Have no ribosomes
(c) Are found only in animals
(d) Have their own DNA and can replicate independently

10. The resolution of a light microscope is approximately:

(a) 200 nm
(b) 2 nm
(c) 0.2 nm
(d) 20 nm

11. The resolution of an electron microscope may be about:

(a) 200 nm
(b) 0.2 nm
(c) 20 nm
(d) 2 µm

12. Which microscope can be used to observe living specimens?

(a) Electron microscope
(b) Both equally
(c) Neither
(d) Light microscope

13. Total magnification of a microscope is calculated by:

(a) Ocular magnification × objective magnification
(b) Objective only
(c) Ocular only
(d) Ocular + objective

14. The cell wall of plants is primarily composed of:

(a) Cellulose
(b) Peptidoglycan
(c) Chitin
(d) Silica

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15. The middle lamella between plant cells is rich in:

(a) Peptidoglycan
(b) Calcium and magnesium pectates
(c) Lignin
(d) Chitin

16. Secondary cell walls are made rigid by:

(a) Chitin
(b) Glycogen
(c) Pectin only
(d) Lignin

17. Cytoplasmic channels connecting adjacent plant cells are called:

(a) Gap junctions
(b) Desmosomes
(c) Plasmodesmata
(d) Tight junctions

18. The fluid mosaic model of the plasma membrane was proposed by:

(a) Watson and Crick
(b) Hooke and Brown
(c) Schleiden and Schwann
(d) Singer and Nicolson

19. The basic framework of the plasma membrane is a:

(a) Phospholipid bilayer
(b) Chitin layer
(c) Cellulose layer
(d) Protein monolayer

20. In the plasma membrane, the hydrophobic tails of phospholipids face:

(a) Outward toward water
(b) Only the nucleus
(c) Inward
(d) Only the cytoplasm

21. Cholesterol in eukaryotic plasma membranes helps to:

(a) Synthesize proteins
(b) Store genetic information
(c) Maintain fluidity at low temperatures
(d) Form cell walls

22. Membrane proteins that form hydrophilic passages are called:

(a) Histones
(b) Ribosomes
(c) Channel proteins
(d) Peripheral proteins only

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23. Aquaporins facilitate the transport of:

(a) Water
(b) Amino acids
(c) DNA
(d) Glucose

24. The nuclear envelope is continuous with the:

(a) Endoplasmic reticulum
(b) Golgi apparatus
(c) Mitochondrion
(d) Lysosome

25. Nuclear pores allow the movement of:

(a) Ribosomes only
(b) Lipids only
(c) RNA and proteins
(d) DNA only

26. Ribosomal subunits are assembled in the:

(a) Peroxisome
(b) Nucleolus
(c) Golgi apparatus
(d) Lysosome

27. Human diploid cells contain how many chromosomes?

(a) 46
(b) 23
(c) 48
(d) 44

28. Rough endoplasmic reticulum is mainly involved in:

(a) Lipid synthesis only
(b) Protein synthesis and processing
(c) Detoxification only
(d) ATP synthesis

29. Smooth endoplasmic reticulum is mainly involved in:

(a) Lipid synthesis and detoxification
(b) DNA replication
(c) Photosynthesis
(d) Protein synthesis

30. In muscle cells, smooth ER stores and releases:

(a) Oxygen
(b) Sodium ions only
(c) Glucose
(d) Calcium ions

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31. Ribosomes are composed of:

(a) RNA only
(b) rRNA and protein
(c) DNA and lipid
(d) Cellulose and protein

32. Eukaryotic cytoplasmic ribosomes are:

(a) 70S
(b) 60S
(c) 40S
(d) 80S

33. Prokaryotic ribosomes are:

(a) 40S
(b) 70S
(c) 80S
(d) 60S

34. Proteins synthesized by free ribosomes are generally used:

(a) Only for secretion
(b) Only for the cell wall
(c) Inside the cell
(d) Only for lysosomes

35. Mitochondria are bounded by:

(a) Two membranes
(b) No membrane
(c) Three membranes
(d) One membrane

36. The infoldings of the inner mitochondrial membrane are called:

(a) Cisternae
(b) Grana
(c) Thylakoids
(d) Cristae

37. F1 particles in mitochondria are actually:

(a) ATP synthase
(b) Ribosomes
(c) Lysosomes
(d) DNA polymerase

38. Mitochondria contain their own:

(a) No genetic material
(b) RNA only
(c) Linear DNA and 80S ribosomes
(d) Circular DNA and 70S ribosomes

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39. The Krebs cycle occurs in the:

(a) Cytoplasm
(b) Nucleus
(c) Thylakoid
(d) Mitochondrial matrix

40. Chloroplasts are the site of:

(a) Photosynthesis
(b) Lipid storage
(c) Protein synthesis only
(d) Cellular respiration only

41. The fluid inside chloroplasts is called:

(a) Matrix
(b) Stroma
(c) Nucleoplasm
(d) Cytosol

42. Stacks of thylakoids in chloroplasts are called:

(a) Grana
(b) Cristae
(c) Lamellae only
(d) Cisternae

43. Chlorophyll molecules are embedded in the:

(a) Nuclear envelope
(b) Thylakoid membranes
(c) Outer membrane
(d) Stroma

44. The Golgi apparatus consists of stacked:

(a) Cisternae
(b) Grana
(c) Cristae
(d) Ribosomes

45. The face of the Golgi apparatus near the ER is called the:

(a) Binding face
(b) Cis face
(c) Catalytic face
(d) Trans face

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46. The Golgi apparatus modifies proteins and lipids and packages them for:

(a) Glycolysis
(b) Secretion or transport
(c) DNA replication
(d) Photosynthesis

47. Lysosomes contain:

(a) Cellulose
(b) Photosynthetic pigments
(c) DNA only
(d) Hydrolytic enzymes

48. Lysosomal enzymes work best at a pH of about:

(a) 4.8
(b) 8.0
(c) 7.4
(d) 2.0

49. The breakdown of worn-out organelles by lysosomes is called:

(a) Phagocytosis
(b) Pinocytosis
(c) Autophagy
(d) Exocytosis

50. Tay-Sachs disease is caused by the absence of a lysosomal enzyme that digests:

(a) Lipids
(b) Nucleic acids
(c) Proteins
(d) Carbohydrates

51. Peroxisomes contain enzymes that break down:

(a) Cellulose
(b) DNA
(c) Glucose
(d) Hydrogen peroxide

52. Glyoxysomes are found in:

(a) Animal cells only
(b) Bacteria only
(c) Plant cells
(d) Viruses

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53. Glyoxysomes are abundant in:

(a) Muscle cells
(b) Red blood cells
(c) Nerve cells
(d) Lipid-rich seeds

54. The membrane surrounding the plant vacuole is called:

(a) Tonoplast
(b) Plasmalemma only
(c) Sarcolemma
(d) Pericardium

55. The fluid inside the vacuole is called:

(a) Cell sap
(b) Stroma
(c) Cytosol
(d) Matrix

56. The central vacuole of plant cells helps maintain:

(a) Photosynthesis only
(b) Blood pressure
(c) Osmosis only
(d) Turgor pressure

57. Centrioles are made of:

(a) Nine doublets of microtubules
(b) Nine triplets of microtubules
(c) Actin filaments
(d) Intermediate filaments

58. Centrioles are absent in:

(a) Animal cells
(b) Higher plant cells
(c) Fungi only
(d) Protists

59. Microfilaments are composed of:

(a) Tubulin
(b) Vimentin
(c) Keratin
(d) Actin

60. Microtubules are composed of:

(a) Actin
(b) Collagen
(c) Myosin
(d) Tubulin

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61. Intermediate filaments in skin cells are made of:

(a) Keratin
(b) Tubulin
(c) Actin
(d) Myosin

62. The core of eukaryotic cilia and flagella is called the:

(a) Centrosome
(b) Sarcomere
(c) Axoneme
(d) Nucleosome

63. Eukaryotic cilia and flagella have a microtubule arrangement of:

(a) 8+2
(b) 9+0
(c) 9+3
(d) 9+2

64. Prokaryotic cells differ from eukaryotic cells in that they lack:

(a) A plasma membrane
(b) Ribosomes
(c) DNA
(d) A true nucleus

65. Prokaryotic DNA is usually:

(a) Linear with histones
(b) Circular and without histones
(c) RNA only
(d) Enclosed in a nuclear envelope

66. Cell signalling begins when a ligand binds to a:

(a) Lysosome
(b) Vacuole
(c) Receptor
(d) Ribosome

67. Second messengers in cell signalling include:

(a) Glucose and starch
(b) DNA and RNA
(c) cAMP and calcium ions
(d) Actin and myosin

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68. Protein/peptide hormones cannot cross the plasma membrane because they are:

(a) Water-soluble
(b) Steroids
(c) Gases
(d) Lipid-soluble

69. Steroid hormones bind to receptors located:

(a) In lysosomes
(b) Only on the cell surface
(c) In the cell wall
(d) Inside the cell

70. The movement of molecules from higher to lower concentration without energy is:

(a) Exocytosis
(b) Active transport
(c) Endocytosis
(d) Diffusion

71. Facilitated diffusion requires:

(a) Transport proteins
(b) Vesicles
(c) Ribosomes
(d) ATP directly

72. Osmosis is the diffusion of:

(a) Oxygen
(b) Amino acids
(c) Glucose
(d) Water

73. Water moves from a hypotonic solution to a:

(a) More dilute solution
(b) Isotonic solution only
(c) Solution with less solute
(d) Hypertonic solution

74. Active transport moves substances:

(a) Against the concentration gradient using energy
(b) Along the gradient without energy
(c) Only by vesicles
(d) Only through channels

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75. The sodium-potassium pump moves:

(a) Na+ in and K+ out
(b) Both ions out
(c) Both ions in
(d) Na+ out and K+ into the cell

76. Phagocytosis is the uptake of:

(a) Liquid droplets
(b) Gases only
(c) Ions only
(d) Solid material

77. Pinocytosis is the uptake of:

(a) DNA only
(b) Solid particles
(c) Liquid droplets
(d) Proteins only

78. Receptor-mediated endocytosis is used by liver cells to take up:

(a) Water only
(b) LDL cholesterol
(c) Oxygen
(d) Glucose only

79. Exocytosis is used to:

(a) Take in solids
(b) Export bulky materials from the cell
(c) Synthesize ATP
(d) Take in liquids

80. Totipotent stem cells can differentiate into:

(a) All possible cell types
(b) Only blood cells
(c) Only muscle cells
(d) Only nerve cells

81. Pluripotent stem cells can form:

(a) Only skin cells
(b) Only blood cells
(c) Almost any cell type
(d) Only one cell type

82. Hematopoietic stem cells are:

(a) Multipotent
(b) Totipotent
(c) Unipotent
(d) Acellular

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83. Embryonic stem cells are derived from the:

(a) Adult bone marrow only
(b) Liver only
(c) Inner cell mass of the blastocyst
(d) Skin only

84. Induced pluripotent stem cells are produced by:

(a) Fertilization
(b) Photosynthesis
(c) Binary fission
(d) Reprogramming adult somatic cells

85. Stem cells are useful in regenerative medicine because they can:

(a) Produce antibiotics
(b) Cause disease only
(c) Form viruses
(d) Repair or replace damaged tissues

86. Which organelle is responsible for detoxification of alcohol in liver cells?

(a) Nucleolus
(b) Peroxisome
(c) Centriole
(d) Ribosome

87. The plasma membrane is described as selectively permeable because it:

(a) Allows some substances to pass while restricting others
(b) Blocks all substances
(c) Allows all substances freely
(d) Is made of cellulose

88. Which of the following is found in both prokaryotic and eukaryotic cells?

(a) Mitochondria
(b) Golgi apparatus
(c) Nucleus
(d) Ribosomes

89. The presence of histones is characteristic of:

(a) Viral RNA only
(b) Plasmids only
(c) Prokaryotic DNA only
(d) Eukaryotic DNA

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90. Which process releases materials from the cell by vesicle fusion with the plasma membrane?

(a) Diffusion
(b) Endocytosis
(c) Osmosis
(d) Exocytosis

91. A plant cell placed in a hypertonic solution will undergo:

(a) No change
(b) Turgor increase
(c) Plasmolysis
(d) Deplasmolysis

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ANSWER KEY WITH BRIEF REASONS

1. (b) Cell — Cell theory states that cells are the basic units of life.

2. (a) Robert Hooke — Hooke observed cork compartments in 1665 and called them cellulae.

3. (a) Anton van Leeuwenhoek — Leeuwenhoek observed animalcules in pond water.

4. (a) Robert Brown — Robert Brown discovered the nucleus in 1831.

5. (b) Matthias Schleiden — Schleiden concluded plants are made of cells.

6. (b) Theodor Schwann — Schwann extended cell theory to animals.

7. (d) Rudolf Virchow — Virchow stated omnis cellula e cellula.

8. (c) Viruses — Viruses are acellular and require a host to replicate.

9. (d) Have their own DNA and can replicate independently — Their semi-autonomous nature suggests endosymbiotic origin.

10. (a) 200 nm — Light microscopes resolve about 200 nm.

11. (b) 0.2 nm — Electron microscopes have much higher resolution.

12. (d) Light microscope — Electron microscopy requires dead/dried specimens.

13. (a) Ocular magnification × objective magnification — Total magnification is the product of lens magnifications.

14. (a) Cellulose — Plant walls contain cellulose microfibrils.

15. (b) Calcium and magnesium pectates — Middle lamella contains pectates and holds adjacent walls.

16. (d) Lignin — Lignin strengthens secondary walls.

17. (c) Plasmodesmata — Plasmodesmata allow intercellular communication.

18. (d) Singer and Nicolson — Singer and Nicolson proposed the model in 1972.

19. (a) Phospholipid bilayer — Membranes are built on a phospholipid bilayer.

20. (c) Inward — Hydrophobic tails avoid water and face inward.

21. (c) Maintain fluidity at low temperatures — Cholesterol modulates membrane fluidity.

22. (c) Channel proteins — Channel proteins allow selective passage of ions/molecules.

47. (d) Hydrolytic enzymes — Lysosomes digest macromolecules.

48. (a) 4.8 — Lysosomes maintain an acidic interior.

49. (c) Autophagy — Autophagy digests the cell’s own components.

50. (a) Lipids — Lipid accumulation damages nerve cells.

51. (d) Hydrogen peroxide — Catalase in peroxisomes detoxifies H_2O_2.

52. (c) Plant cells — Glyoxysomes convert lipids to carbohydrates in seeds.

53. (d) Lipid-rich seeds — They provide energy during seed germination.

54. (a) Tonoplast — Tonoplast encloses the vacuole.

55. (a) Cell sap — Cell sap differs from cytoplasm.

56. (d) Turgor pressure — Turgor supports plant rigidity.

57. (b) Nine triplets of microtubules — Centrioles have a 9+0 triplet arrangement.

58. (b) Higher plant cells — Plants and fungi lack centrioles/basal bodies.

59. (d) Actin — Microfilaments are actin rods.

60. (d) Tubulin — Microtubules are made of alpha/beta tubulin.

61. (a) Keratin — Keratin strengthens skin cells.

62. (c) Axoneme — Axoneme has the 9+2 microtubule pattern.

63. (d) 9+2 — Nine doublets surround two central microtubules.

64. (d) A true nucleus — Prokaryotes have a nucleoid, not a membrane-bound nucleus.

65. (b) Circular and without histones — Bacterial chromosomes are circular and naked.

66. (c) Receptor — Reception is the first step of signalling.

67. (c) cAMP and calcium ions — cAMP, Ca^{2+} and IP3 relay signals.

68. (a) Water-soluble — They bind surface receptors instead.

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23. (a) Water — Aquaporins are water channels.

24. (a) Endoplasmic reticulum — The outer nuclear membrane connects with ER.

25. (c) RNA and proteins — Nuclear pores regulate nucleocytoplasmic transport.

26. (b) Nucleolus — The nucleolus produces rRNA and assembles ribosomal subunits.

27. (a) 46 — Human somatic cells have 46 chromosomes.

28. (b) Protein synthesis and processing — RER has ribosomes and processes proteins.

29. (a) Lipid synthesis and detoxification — SER lacks ribosomes and makes lipids/detoxifies.

30. (d) Calcium ions — Calcium release triggers muscle contraction.

31. (b) rRNA and protein — Ribosomes are about 60% rRNA and 40% protein.

32. (d) 80S — Eukaryotic cytoplasmic ribosomes are 80S.

33. (b) 70S — Prokaryotes have 70S ribosomes.

34. (c) Inside the cell — Free ribosomes make cytosolic proteins.

35. (a) Two membranes — Mitochondria have outer and inner membranes.

36. (d) Cristae — Cristae increase surface area.

37. (a) ATP synthase — F1 particles synthesize ATP.

38. (d) Circular DNA and 70S ribosomes — Mitochondria are semi-autonomous.

39. (d) Mitochondrial matrix — Matrix enzymes catalyse the Krebs cycle.

40. (a) Photosynthesis — Chloroplasts convert light energy into chemical energy.

41. (b) Stroma — Stroma contains enzymes for the Calvin cycle.

42. (a) Grana — Grana are thylakoid stacks.

43. (b) Thylakoid membranes — Light absorption occurs in thylakoid membranes.

44. (a) Cisternae — Golgi stacks are made of cisternae.

45. (b) Cis face — Cis face receives vesicles from ER.

69. (d) Inside the cell — Steroids are lipophilic and enter cells.

70. (d) Diffusion — Diffusion is passive.

71. (a) Transport proteins — Carrier/channel proteins assist passive movement.

72. (d) Water — Osmosis is passive water movement across a selectively permeable membrane.

73. (d) Hypertonic solution — Water follows its concentration gradient.

74. (a) Against the concentration gradient using energy — Active transport requires ATP.

75. (d) Na+ out and K+ into the cell — It maintains ion gradients in nerve cells.

76. (d) Solid material — Phagocytosis is cell eating.

77. (c) Liquid droplets — Pinocytosis is cell drinking.

78. (b) LDL cholesterol — LDL receptors mediate cholesterol uptake.

79. (b) Export bulky materials from the cell — Vesicles fuse with the membrane to release contents.

80. (a) All possible cell types — Totipotent cells, such as the zygote, have full potential.

81. (c) Almost any cell type — Pluripotent cells have broad but not full potential.

82. (a) Multipotent — They form related blood cell types.

83. (c) Inner cell mass of the blastocyst — ESCs come from early embryos.

84. (d) Reprogramming adult somatic cells — iPSCs are generated using transcription factors.

85. (d) Repair or replace damaged tissues — They can generate specialized cells for therapy.

86. (b) Peroxisome — Peroxisomes oxidize harmful compounds.

87. (a) Allows some substances to pass while restricting others — Membranes regulate transport.

88. (d) Ribosomes — Ribosomes are universal.

89. (d) Eukaryotic DNA — Eukaryotic chromosomes use histones.

90. (d) Exocytosis — Exocytosis exports secretions.

91. (c) Plasmolysis — Water leaves the cell, shrinking the protoplast.

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46. (b) Secretion or transport — Golgi is the shipping department of the cell.

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